BioAuth Project Backlog

TED University CMPE 492 Senior Design Project II
Group 1 | Planning revision 1 | 18 October 2026

Planning proposal. Use one project backlog to coordinate the four members. Each active item has one proposed accountable owner and a second proposed reviewer. The team must agree the assignments and internal targets before treating them as commitments.

The original 15 October v1 remains unchanged in the archive. This revision preserves B01-B29, adds the early feasibility check B30, and separates documented progress from unconfirmed implementation and submission status.

Proposed responsibilities

Member	Primary planning area
Enes Taşçı	API, platform integration and packaging
Esra Eda Kılıç	rPPG, capture quality and evaluation data
Batuhan Mert Korkmaz	Face verification, Visual PAD and fusion
Ömer Taylan Alemdar	Capture UI, project coordination and test planning

The owner coordinates delivery; the reviewer checks the acceptance evidence. Everyone contributes to integration, reporting and the final presentation. Item counts are not effort estimates or individual contribution grades.

Course delivery dates

Course date	Required snapshot and deliverable
16 Oct 2026	LLD v1 and backlog; actual submission record needs confirmation
23 Oct 2026	LLD v2 following advisor feedback and updated backlog
20 Nov 2026	Test Plan and updated backlog
31 Dec 2026	Final Report and updated backlog
04-17 Jan 2027	Presentation and demo; exact date TBA

The dated Fall 2026-2027 table on syllabus page 7 sets the course deadlines at 23:59. Task targets below are proposed internal checkpoints unless they coincide with those deliverables. The 18 October revision does not prove publication or submission by 16 October.

Backlog items

P0: core correctness or course delivery. P1: integration, evaluation or packaging. P2: deferred extension. Status records the evidence available on 18 October; it does not establish individual contribution.

Snapshot: 25 planned; 1 review; 1 needs confirmation; 1 complete; 2 deferred. No implementation completion percentage is inferred.

B01 Review the six LLD design decisions

P0 | Review | Target: 23 Oct 2026

Proposed owner: Ömer Taylan Alemdar | Proposed reviewer: Esra Eda Kılıç
Dependencies: None | Design and delivery

Done when Record the team's review of mandatory usable rPPG, nullable evidence, review guards, camera-derived templates, the C-Q comparison and outcome denominators. Resolve advisor questions in the revision log.

Trace LLD 1.1.3 and 3.15.2; ADR-01 to ADR-06

Record LLD v1 dated 16 October is available. Team/advisor acceptance is not recorded in this snapshot.

B02 Verify course submissions and backlog publication

P0 | Needs confirmation | Target: 23 Oct 2026

Proposed owner: Enes Taşçı | Proposed reviewer: Ömer Taylan Alemdar
Dependencies: B01 | Design and delivery

Done when Check the 16 October delivery record and complete the 23 October delivery checklist: LMS, required email recipients, coordinator form and public report/backlog links. Record actual submission and publication dates separately.

Trace Syllabus pp. 1, 4 and 7

Record LLD v1 is publicly available. LMS/email/form submission is unconfirmed; a preparation date is not proof of timely submission.

B03 Revise LLD with advisor feedback

P0 | Planned | Target: 23 Oct 2026

Proposed owner: Ömer Taylan Alemdar | Proposed reviewer: Batuhan Mert Korkmaz
Dependencies: B01, B30 | Design and delivery

Done when Produce LLD v2 with a dated change log. Resolve the web-first/native scope and inherited timing feasibility with the advisor; preserve the mandatory-rPPG safety rule.

Trace LLD 3.14 and 3.15.2; syllabus p. 7

B04 Create a reproducible service skeleton

P0 | Planned | Target: 23 Oct 2026

Proposed owner: Enes Taşçı | Proposed reviewer: Batuhan Mert Korkmaz
Dependencies: B01 | Platform and security

Done when A clean checkout starts the Core facade, API health route, database migration and experiment runner with pinned dependencies. Document the service boundaries and run a smoke check.

Trace LLD 2 and 3.14; V2 19 and 20

B05 Define evidence and decision contracts

P0 | Planned | Target: 23 Oct 2026

Proposed owner: Batuhan Mert Korkmaz | Proposed reviewer: Enes Taşçı
Dependencies: B01 | Evidence and fusion

Done when Implement matching Python/TypeScript schemas for status, nullable values, score direction, reason codes and versioned snapshots. Contract tests distinguish unusable measurements, real zero values and technical errors.

Trace LLD 1.2 and 3.1; ADR-01; TC-FUSION-01 to 08

B06 Qualify model artifacts and adapters

P0 | Planned | Target: 23 Oct 2026

Proposed owner: Batuhan Mert Korkmaz | Proposed reviewer: Esra Eda Kılıç
Dependencies: B01 | Evidence and fusion

Done when Record exact embedding/PAD weights, checksums, preprocessing, dimensions, score semantics and permitted use. Missing or incompatible manifests fail readiness; do not redistribute unapproved artifacts.

Trace LLD 3.4.2 and 3.5.4; TC-MATCH-01; TC-PAD-01

B07 Implement session lifecycle and access control

P0 | Planned | Target: 30 Oct 2026

Proposed owner: Enes Taşçı | Proposed reviewer: Ömer Taylan Alemdar
Dependencies: B04, B05 | Platform and security

Done when Enforce session ownership, expiry, allowed state transitions, cancellation, idempotent finalization and retry limits. Terminal sessions reject later writes and cross-user access.

Trace LLD 3.2; TC-AUDIT-01; TC-DASH-01

B08 Build browser capture and bounded transport

P0 | Planned | Target: 30 Oct 2026

Proposed owner: Ömer Taylan Alemdar | Proposed reviewer: Enes Taşçı
Dependencies: B05, B07 | Capture and interfaces

Done when Send versioned WSS packets with monotonic offsets, sequence and dimensions. Measure delivered FPS, gaps, backpressure and dropped frames; stop camera tracks and release buffers on every terminal path.

Trace LLD 3.3.3 and 3.11; TC-INPUT-01; TC-QUALITY-02

B09 Implement landmarks ROI and capture quality

P0 | Planned | Target: 30 Oct 2026

Proposed owner: Esra Eda Kılıç | Proposed reviewer: Batuhan Mert Korkmaz
Dependencies: B05, B06 | Evidence and fusion

Done when Track one face with stable forehead/cheek ROIs and timed RGB traces. Detect invalid bounds, face changes, blur, exposure and movement. Test first with controlled fixtures, then browser captures from B08.

Trace LLD 3.3; FR-01 to FR-03 and FR-08; TC-FACE-01; TC-ROI-01

B10 Implement face comparison and template storage

P0 | Planned | Target: 06 Nov 2026

Proposed owner: Batuhan Mert Korkmaz | Proposed reviewer: Enes Taşçı
Dependencies: B05, B06, B09 | Evidence and fusion

Done when Compare compatible embeddings from quality-selected frames. Encrypt candidate/active templates, preserve camera-capture provenance and model versions, and prevent candidate templates from authorizing returning users.

Trace LLD 3.4; ADR-06; TC-MATCH-01; TC-ENROLL-01

B11 Integrate Visual PAD and temporal aggregation

P0 | Planned | Target: 06 Nov 2026

Proposed owner: Batuhan Mert Korkmaz | Proposed reviewer: Esra Eda Kılıç
Dependencies: B06, B09 | Evidence and fusion

Done when Run the qualified PAD adapter with exact preprocessing and deterministic sampling. Report median, lower quantile, spread and eligible counts. Separate inference errors and unusable crops from qualified attack evidence.

Trace LLD 3.5; V2-FR-11; TC-PAD-01 to 03

B12 Implement POS CHROM and signal qualification

P0 | Planned | Target: 06 Nov 2026

Proposed owner: Esra Eda Kılıç | Proposed reviewer: Batuhan Mert Korkmaz
Dependencies: B05, B09, B30 | Evidence and fusion

Done when Validate timed RGB, POS/CHROM, the configured 0.7-4 Hz filter and Welch spectrum with known signals and permitted reference data. Qualify duration, SNR and agreement; missing/unusable BPM remains null.

Trace LLD 3.6; FR-04 to FR-06; TC-RPPG-01 to 04

B13 Qualify transported physiological evidence

P0 | Planned | Target: 13 Nov 2026

Proposed owner: Esra Eda Kılıç | Proposed reviewer: Ömer Taylan Alemdar
Dependencies: B08, B12, B30 | Evaluation and release

Done when Compare the browser transport with a lossless research capture. Measure usable duration, timestamp gaps and compression effects before freezing minimum-rPPG criteria; record any proposed timing/profile revision.

Trace LLD 3.3.3 and 3.6.4; TC-RPPG-02 to 04; TC-QUALITY-02

B14 Enforce enrollment and manual review guards

P0 | Planned | Target: 19 Nov 2026

Proposed owner: Enes Taşçı | Proposed reviewer: Batuhan Mert Korkmaz
Dependencies: B07, B10, B11, B12, B15 | Platform and security

Done when Activate an account/template only after biometric ACCEPT and an authorized review bound to the same candidate and snapshot. Test stale approvals, duplicate actions and prohibited overrides of REJECT/RETRY.

Trace LLD 3.8; ADR-02 and ADR-06; TC-ENROLL-01 to 04

B15 Implement the common safety gate and L1

P0 | Planned | Target: 13 Nov 2026

Proposed owner: Batuhan Mert Korkmaz | Proposed reviewer: Esra Eda Kılıç
Dependencies: B10, B11, B12 | Evidence and fusion

Done when Test reliable-rejection precedence, technical failures, missing evidence and mandatory usable rPPG. Every actionable acceptance passes the shared gate; L1 uses a versioned validation-calibrated fixed policy.

Trace LLD 3.7.2 and 3.7.3; ADR-05; TC-FUSION-01 to 08

B16 Implement quality-only C-Q and attack-aware L2

P0 | Planned | Target: 19 Nov 2026

Proposed owner: Batuhan Mert Korkmaz | Proposed reviewer: Esra Eda Kılıç
Dependencies: B15 | Evidence and fusion

Done when Compare C-Q and L2 on identical snapshots and common gates. Unit-test PAD temporal instability, physiology disagreement and PAD suspicion with pulse. Record pattern rule versions and their cost in Retry/coverage.

Trace LLD 3.7.3 to 3.7.5; ADR-03; TC-FUSION-05 and 08

B17 Bind verification to one sensitive action

P0 | Planned | Target: 27 Nov 2026

Proposed owner: Enes Taşçı | Proposed reviewer: Ömer Taylan Alemdar
Dependencies: B07, B14, B15 | Platform and security

Done when Bind successful verification to its owner, intended payout-destination change, digest and expiry. Atomically consume it once; test changed actions, expired/replayed results and concurrent double execution.

Trace LLD 3.9; V2-FR-13; TC-DEMO-01

B18 Connect user guidance and authorized admin views

P1 | Planned | Target: 27 Nov 2026

Proposed owner: Ömer Taylan Alemdar | Proposed reviewer: Enes Taşçı
Dependencies: B08, B14, B16 | Capture and interfaces

Done when Render real session events, quality guidance, decisions and review state. Separate ordinary user/admin fields from consented research diagnostics, and verify authorization before event serialization.

Trace LLD 3.10 and 3.11; V2-FR-15; TC-DASH-01 to 03

B19 Prepare permitted data and consent protocol

P0 | Planned | Target: 30 Oct 2026

Proposed owner: Esra Eda Kılıç | Proposed reviewer: Ömer Taylan Alemdar
Dependencies: B01 | Evaluation and release

Done when Record dataset permissions, consent scope, storage/deletion rules and a fallback if access is delayed. Distinguish PAD-labelled data, physiological reference data and a paired multimodal corpus; none substitutes for the others.

Trace LLD 3.12 and 3.13; V2 17 and 18

B20 Write the Test Plan and split manifest

P0 | Planned | Target: 20 Nov 2026

Proposed owner: Ömer Taylan Alemdar | Proposed reviewer: Esra Eda Kılıç
Dependencies: B05, B19 | Design and delivery

Done when Define test IDs, fixtures/data, expected results, environments and accountable reviewers. Freeze subject/source separation and device holdouts where feasible; keep physical PAD and injection tests separate and specify all outcome denominators.

Trace LLD 3.13 and 3.15; syllabus pp. 3 and 7

B21 Build paired evidence tables and honest metrics

P1 | Planned | Target: 27 Nov 2026

Proposed owner: Esra Eda Kılıç | Proposed reviewer: Batuhan Mert Korkmaz
Dependencies: B11, B12, B16, B20 | Evaluation and release

Done when Run C, C-Q and D on the same evidence and splits, with A/B limited to diagnostic controls. Count ACCEPT, REJECT, RETRY and technical outcomes over initiated attempts; report coverage and capture-to-decision latency separately from cached-policy cost.

Trace LLD 3.13; ADR-04; TC-AUDIT-01; V2 21

B22 Assess the L3 advisory experiment

P1 | Planned | Target: 04 Dec 2026

Proposed owner: Esra Eda Kılıç | Proposed reviewer: Batuhan Mert Korkmaz
Dependencies: B21 | Evaluation and release

Done when If the paired training data is adequate, fit regularized logistic regression with training-only preprocessing and validation-only calibration. Preserve all safety holds; otherwise record L3 as unavailable and report the limitation. XGBoost remains optional.

Trace LLD 3.7.6; V2 9 and 21

B23 Freeze profiles and run the final evaluation

P0 | Planned | Target: 11 Dec 2026

Proposed owner: Esra Eda Kılıç | Proposed reviewer: Ömer Taylan Alemdar
Dependencies: B13, B20, B21, B22 | Evaluation and release

Done when Record the L3 availability decision, then freeze model, capture, policy and calibration manifests before opening final labels. Produce paired policy results, per-species attack outcomes, Retry/coverage, uncertainty and p50/p95 latency without retuning on test results.

Trace LLD 3.13 and 3.14; TC-AUDIT-01

B24 Verify security privacy and failure handling

P0 | Planned | Target: 11 Dec 2026

Proposed owner: Ömer Taylan Alemdar | Proposed reviewer: Enes Taşçı
Dependencies: B14, B17, B18 | Platform and security

Done when Verify cross-user isolation, review override prevention, single consumption, cancellation/expiry cleanup and retention. Confirm ordinary interfaces expose no raw frames, embeddings or unauthorized research diagnostics.

Trace LLD 3.8 to 3.12; TC-ENROLL-01 to 04; TC-DEMO-01; TC-DASH-01 to 03

B25 Align the public project presentation with V2

P1 | Complete | Target: 18 Oct 2026

Proposed owner: Enes Taşçı | Proposed reviewer: Ömer Taylan Alemdar
Dependencies: None | Capture and interfaces

Done when Present the current scope, academic reports, team and evaluation plan; label the earlier scenario console as simulated. This item covers the public presentation, not the operational biometric engine or live admin application.

Trace LLD 3.14.1; public project presentation

Record BioAuth master commit 5381287 and the released project website, reviewed 18 October 2026. Proposed owners do not establish past individual contribution.

B26 Package Core SDK and a clean integration

P1 | Planned | Target: 18 Dec 2026

Proposed owner: Enes Taşçı | Proposed reviewer: Batuhan Mert Korkmaz
Dependencies: B06, B23, B24 | Evaluation and release

Done when Pass a clean-environment build and sample integration with pinned dependencies, migrations, allowed model artifacts and notices. Prepare versioned Python/TypeScript packages without biometric data or secrets; confirm names/rights before registry publication.

Trace LLD 3.14.4; V2-FR-17; TC-PACK-01 to 03

B27 Assemble the final report and team demonstration

P0 | Planned | Target: 31 Dec 2026

Proposed owner: Ömer Taylan Alemdar | Proposed reviewer: Esra Eda Kılıç
Dependencies: B23, B24, B26 | Design and delivery

Done when Report actual architecture, test-ID pass/fail evidence, measured results and limitations. Include all four members in the presentation, rehearse within 25 minutes and prepare the source/demo materials for the January date announced by the course.

Trace Syllabus pp. 3, 4 and 7; LLD 3.15

B28 Explore a dynamic light challenge

P2 | Deferred | Target: Future scope

Proposed owner: Unassigned | Proposed reviewer: Unassigned
Dependencies: B23 | Future scope

Done when Require a separate scope decision, fresh acquisition protocol and tests for interference with rPPG before implementation. Do not infer challenge results from existing recordings.

Trace LLD 3.14.4; V2-FR-16; TC-CHAL-01 to 02

Record Outside the committed first implementation.

B29 Explore native capture or deep rPPG

P2 | Deferred | Target: Future scope

Proposed owner: Unassigned | Proposed reviewer: Unassigned
Dependencies: B23 | Future scope

Done when Consider native capture and deep-rPPG as separate extensions after core evaluation and an explicit capacity/scope review. Keep OCR, NFC and document-authenticity models outside the initial scope.

Trace LLD 3.14.4; V2 4 and 19

Record No active owner or delivery promise.

B30 Check short-window and transport feasibility early

P0 | Planned | Target: 23 Oct 2026

Proposed owner: Esra Eda Kılıç | Proposed reviewer: Enes Taşçı
Dependencies: None | Evidence and fusion

Done when Use a small reproducible signal/capture spike to expose the five-second window, natural frequency resolution, gap and processing-budget risks before full integration. Bring feasibility findings or unresolved risks to LLD v2 review; B13 later performs full qualification.

Trace LLD 3.3.3, 3.6.4 and 3.14.3

Record Added in the 18 October planning revision to surface a critical dependency earlier.

Weekly review and completion evidence

At each weekly meeting, update status, blockers, dependencies and evidence. The owner proposes next steps; the reviewer checks the acceptance criteria and records the review date.

Planned has no completion evidence; Review awaits acceptance; Needs confirmation requires an external record. Complete needs a commit, test result or reviewed artifact. Deferred has no delivery commitment.

Record actual contributors and test IDs separately from proposed owners. At each checkpoint, preserve the dated PDF and task export. Keep submission receipts and public publication dates as separate records; never backdate a snapshot.

Implementation safeguards

Usable rPPG remains mandatory for actionable ACCEPT across L1, C-Q, L2 and L3. A pulse is physiological evidence, not proof of fresh capture. Missing evidence cannot be hidden by weighting, an admin review or a learned model. Camera-derived templates require successful biometric enrollment and a bound approval.

C, C-Q and D must use the same evidence, shared gates and evaluation splits. Report ACCEPT, REJECT, RETRY, technical outcomes, coverage and latency together. Keep final-test labels out of threshold selection. L3 availability depends on adequate data; it cannot be claimed by relabelling an L2 result.

Priority risks

Address short-window and transport feasibility early in B30, then qualify the complete path in B13. Secure data access and consent in B19 before finalizing experiments. Keep core correctness, the Test Plan and controlled evaluation ahead of optional light challenge, native capture and deep-rPPG extensions.

Source and revision record

CMPE 492 syllabus, Fall 2026-2027: backlog requirement on p. 4, example on p. 5, grading on p. 6 and dated delivery table on p. 7. The example is not a required template.

BioAuth Low-Level Design Report, 16 October 2026: ADRs in 1.1.3, implementation contracts in 3.1-3.12, evaluation in 3.13, delivery in 3.14 and traceability in 3.15. This is the detailed implementation baseline.

BioAuth V2 Technical Foundation, Product Architecture and Implementation Roadmap, Design Baseline v2.0, 12 October 2026: research scope in 9, roadmap in 19-20 and paired-evidence experiments in 21. The LLD refines this reference.

BioAuth_Backlog_v1.docx, version 1 dated 15 October 2026: original B01-B29. This revision adds proposed named owners/reviewers, updates dependency targets, separates publication from course submission, records completed presentation work B25, and adds B30.